Coordination-Dependent Hydrogen Adsorption on Co-, Fe-, and Ni-Functionalized Monolayer $CrCl_3$

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Supporting Information Available

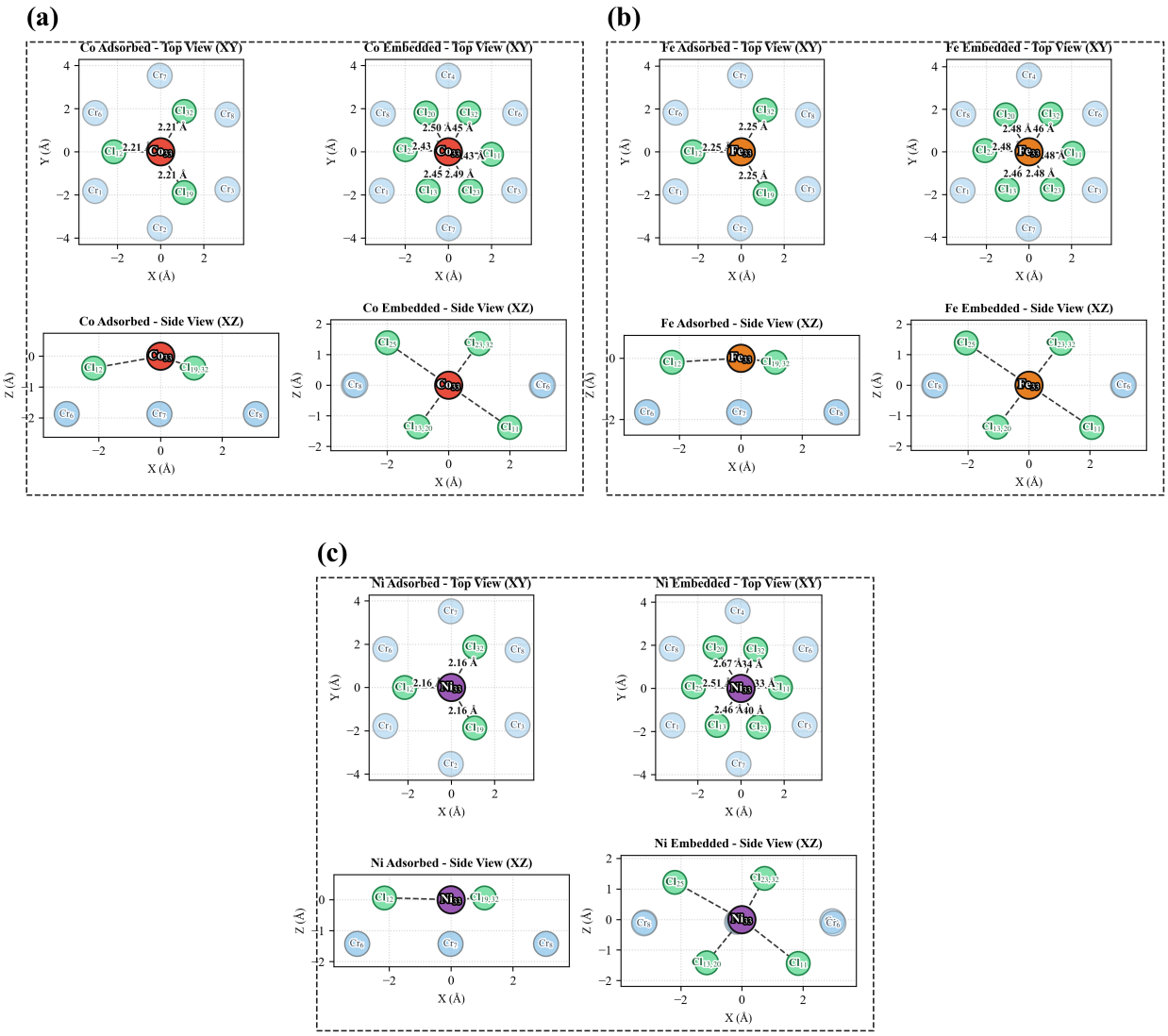


Fig. S1. Schematic top (XY) and side (XZ) views of the optimized local coordination geometries and TM–Cl bond lengths (Å) for (a) Co-, (b) Fe-, and (c) Ni-functionalized monolayer $CrCl_3$ in both adsorbed (hollow site, 3-fold coordinated) and embedded (6-fold octahedrally coordinated) configurations. Light blue, light green, and red/orange/purple spheres represent Cr, Cl, and transition-metal (Co, Fe, Ni) atoms, respectively.

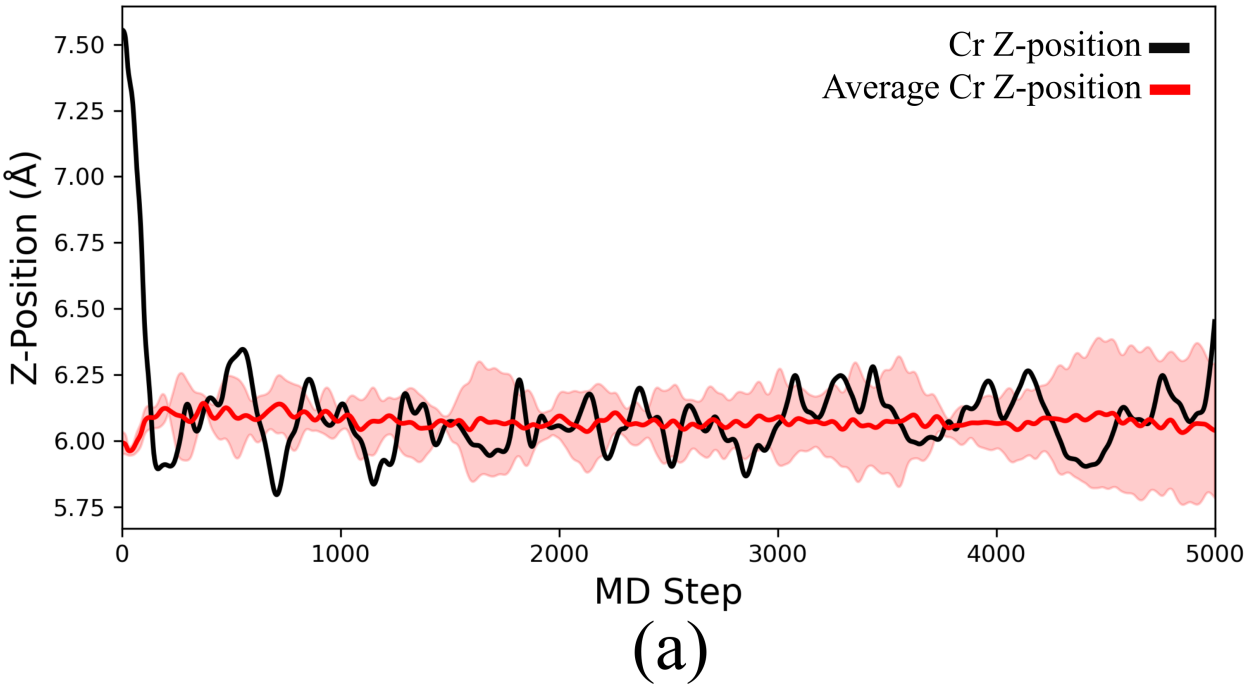


Fig. S2. Time evolution of the absolute z coordinate of the Cr atom initially adsorbed at the hollow site of monolayer $CrCl_3$ during a 5 ps *ab initio* molecular dynamics simulation at 300 K. The black curve represents the instantaneous z coordinate of the adsorbed Cr atom, whereas the red curve denotes the average z coordinate of the Cr atoms belonging to the host monolayer.

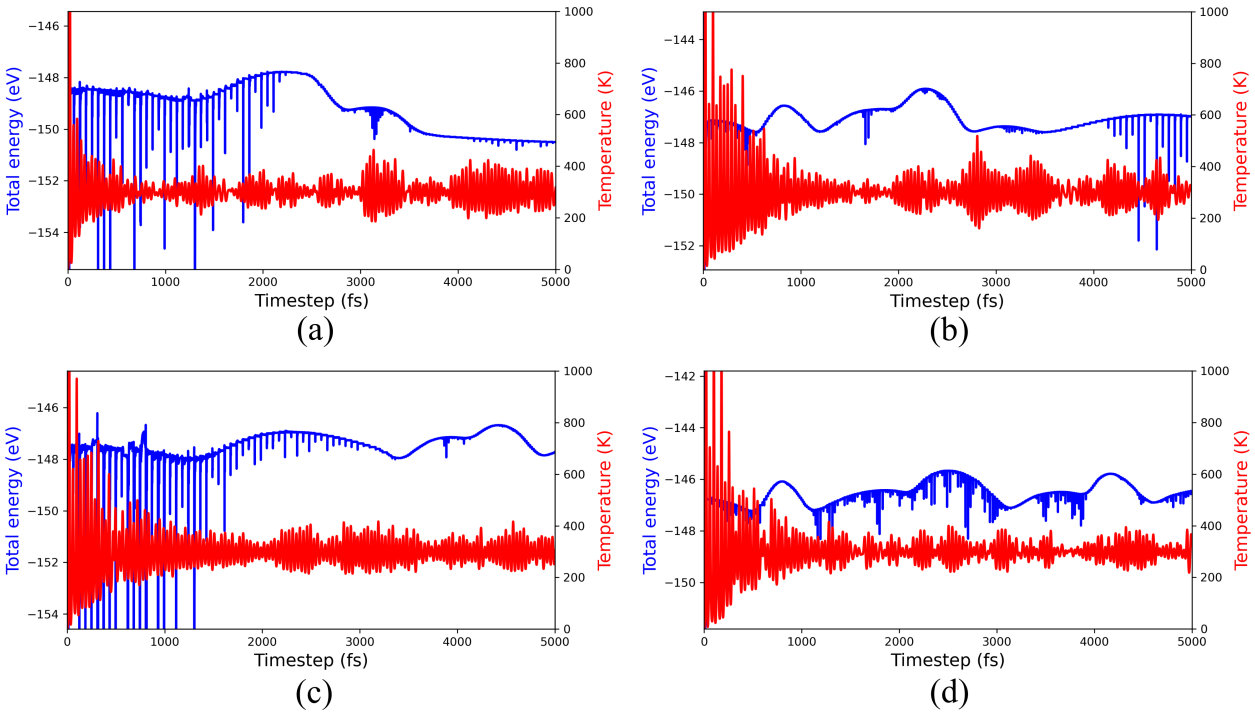


Fig. S3. Time evolution of the total energy (blue curves, left axes) and instantaneous temperature (red curves, right axes) during the 5 ps AIMD simulations at 300 K for monolayer $CrCl_3$ containing an initially surface-adsorbed (a) Cr, (b) Co, (c) Fe, or (d) Ni atom at the hollow site. Following the initial thermostatting transients, the temperatures fluctuate around the target value. The changes in the total-energy profiles accompany the structural rearrangements sampled during the trajectories.

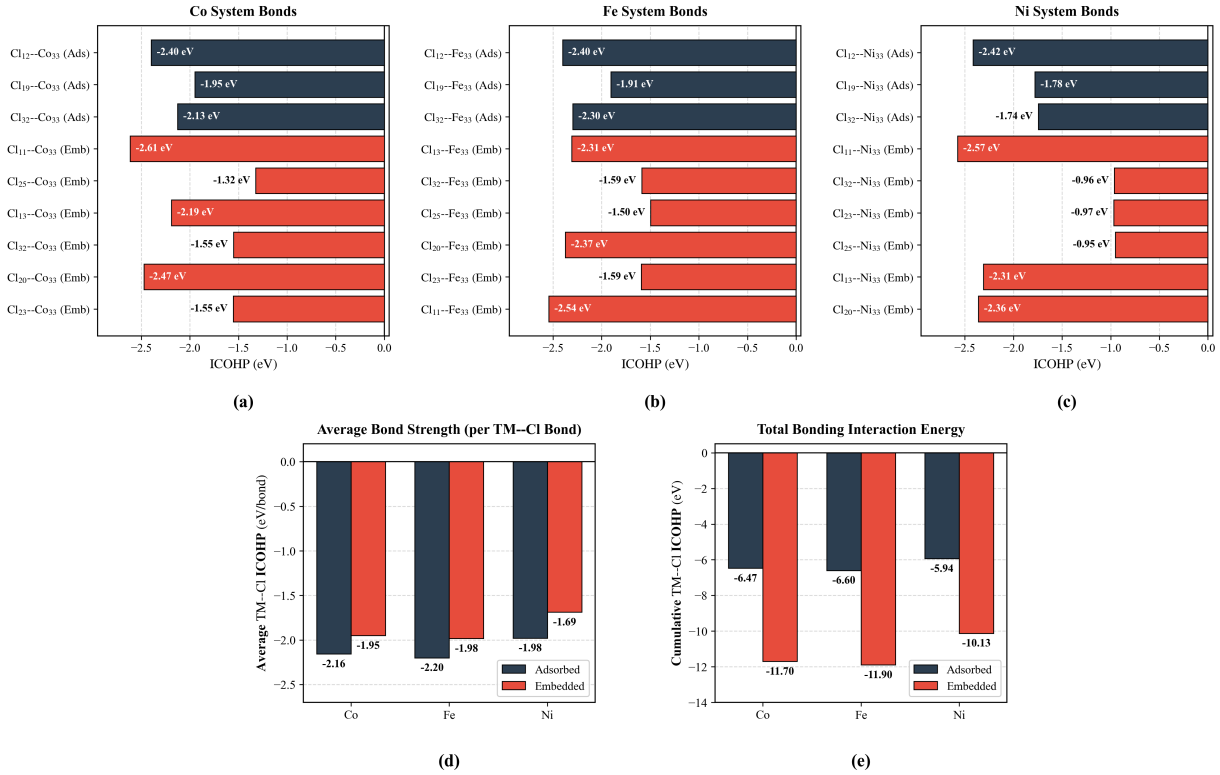


Fig. S4. TM–Cl chemical-bonding analysis and relative stability of surface-adsorbed and embedded transition-metal configurations in monolayer $CrCl_3$ under the multi-site $+U_{\mathrm{all}}$ tier ($U_{\mathrm{Cr}} = U_{\mathrm{TM}} = 3.29$ eV; consistent with Figure 6, with full Pure PBE and $+U_{\mathrm{all}}$ comparisons consolidated in Table S4). (a–c) Individual integrated crystal orbital Hamilton population (ICOHP) values for the Co–Cl, Fe–Cl, and Ni–Cl interactions, respectively. Dark navy bars correspond to the threefold coordinated surface-adsorbed configurations at the S_2 hollow site, whereas red bars correspond to the sixfold coordinated embedded configurations. More negative ICOHP values indicate stronger net bonding interactions. (d) Average ICOHP per TM–Cl bond for the adsorbed and embedded configurations. (e) Cumulative TM–Cl ICOHP obtained by summing the individual ICOHP contributions over the corresponding coordination shell.

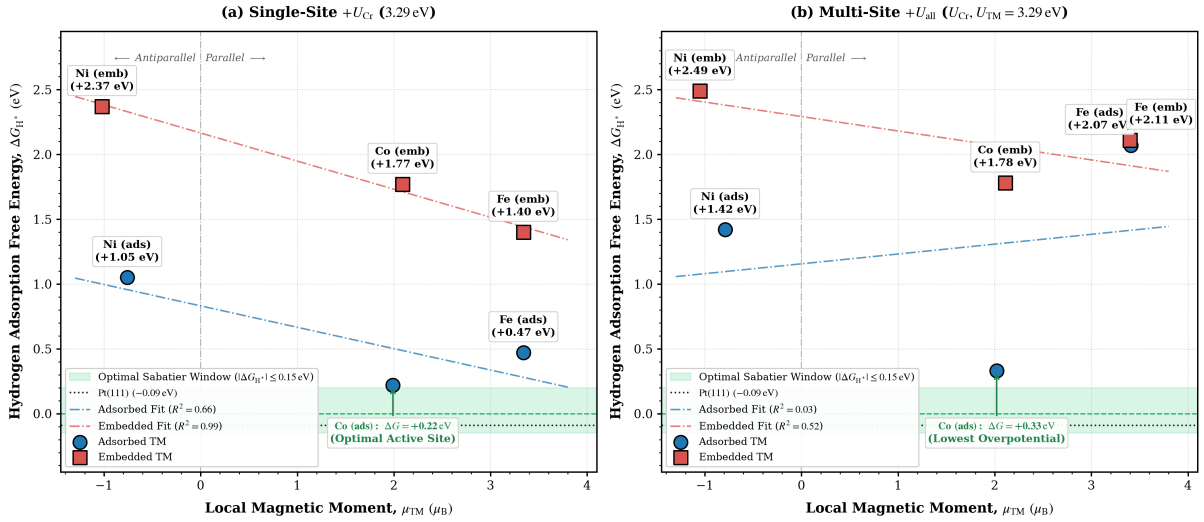


Fig. S5. Relationship between the local magnetic moment of the functionalizing transition-metal atom, μ_{TM}, in the relaxed H-free configurations and the hydrogen adsorption free energy, ΔG_{H^*}, across two systematic Hubbard U tiers: (a) single-site $+U_{Cr}$ (U_{Cr} = 3.29 eV) and (b) multi-site $+U_{all}$ ($U_{Cr} = U_{TM}$ = 3.29 eV). Blue circles and red squares represent the surface-adsorbed (S_2) and embedded transition-metal configurations in monolayer $CrCl_3$, respectively. The vertical dash-dotted line at $\mu_{TM} = 0$ separates antiparallel (Ni) from parallel (Co, Fe) magnetic coupling relative to the ferromagnetic Cr host. Shaded green bands denote the optimal Sabatier catalytic window ($|\Delta G_{H^*}| \leq 0.15$ eV) with the Pt(111) benchmark (−0.09 eV) indicated by dotted horizontal lines. Magnetic moments are expressed in units of the Bohr magneton, μ_B.

Supplementary Tables: Comprehensive Structural, Energetic, and Thermodynamic Benchmarks

Table S1: Multi-Scale Hydrogen Adsorption on Pristine Monolayer $CrCl_3$

Table S1. Comprehensive multi-scale evaluation of hydrogen adsorption on pristine monolayer $CrCl_3$ across three supercell dimensions (1×1, 2×2, and 3×3), three high-symmetry adsorption sites (S1: Top-Cl, S2: Hollow, S3: Top-Cr), and three levels of exchange-correlation dispersion treatment (pure PBE, PBE+D3 zero-damping, and PBE+D3 Becke-Johnson damping). All calculations were performed spin-polarized with $E_{\text{cut}} = 400$ eV and commensurate Monkhorst-Pack k-point grids ($10\times10\times1$, $5\times5\times1$, and $3\times3\times1$, respectively). Adsorption energies are referenced to isolated H_2 gas: $\Delta E_{\text{ads}} = E_{\text{host+H}} - E_{\text{host}} - \frac{1}{2}E(H_2)$. Free energies are evaluated as $\Delta G_{\text{ads}} = \Delta E_{\text{ads}} + \Delta\text{ZPE} - T\Delta S$, where explicit phononic columns (ΔZPE and $-T\Delta S$) are provided for complete vibrational frequency assignment.

Supercell	Site	Coord.	Method / vdW	E_{host} (eV)	$E_{\text{host+H}}$ (eV)	ΔE_{ads} (eV)	ΔZPE^* (eV)	$-T\Delta S^*$ (eV)	ΔG_{ads} (eV)	μ_{tot} (μ_B)
1×1 ($\theta = 1.00$)	S1	Top-Cl	PBE (no vdW)	−39.0717	−40.7734	+1.678	+0.040	+0.200	+1.918	7.00
	S2	Hollow	PBE (no vdW)	−39.0717	−39.9647	+2.487	+0.040	+0.200	+2.727	5.04
	S3	Top-Cr	PBE (no vdW)	−39.0717	−40.8155	+1.636	+0.040	+0.200	+1.876	5.00
	S1	Top-Cl	PBE+D3 (zero)	−39.9841	−41.8232	+1.542	+0.040	+0.200	+1.782	7.00
	S2	Hollow	PBE+D3 (zero)	−39.9841	−40.9868	+2.378	+0.040	+0.200	+2.618	5.51
	S3	Top-Cr	PBE+D3 (zero)	−39.9841	−41.7093	+1.656	+0.040	+0.200	+1.896	5.00
	S1	Top-Cl	PBE+D3 (BJ)	−40.5268	−42.5826	+1.325	+0.040	+0.200	+1.565	7.00
	S2	Hollow	PBE+D3 (BJ)	−40.5268	−41.8481	+2.060	+0.040	+0.200	+2.300	7.00
	S3	Top-Cr	PBE+D3 (BJ)	−40.5268	−42.2405	+1.667	+0.040	+0.200	+1.907	5.00
2×2 ($\theta = 0.25$)	S1	Top-Cl	PBE (no vdW)	−156.2868	−158.0877	+1.579	+0.040	+0.200	+1.819	25.00
	S2	Hollow	PBE (no vdW)	−156.2868	−157.1957	+2.471	+0.040	+0.200	+2.711	23.00
	S3	Top-Cr	PBE (no vdW)	−156.2868	−158.0422	+1.624	+0.040	+0.200	+1.864	23.00
	S1	Top-Cl	PBE+D3 (zero)	−159.9366	−161.8036	+1.514	+0.040	+0.200	+1.754	25.00
	S2	Hollow	PBE+D3 (zero)	−159.9366	−160.9571	+2.361	+0.040	+0.200	+2.601	23.00
	S3	Top-Cr	PBE+D3 (zero)	−159.9366	−161.6926	+1.625	+0.040	+0.200	+1.865	23.00
	S1	Top-Cl	PBE+D3 (BJ)	−162.1074	−164.2008	+1.288	+0.040	+0.200	+1.528	25.00
	S2	Hollow	PBE+D3 (BJ)	−162.1074	−163.1641	+2.324	+0.040	+0.200	+2.564	23.01
	S3	Top-Cr	PBE+D3 (BJ)	−162.1074	−163.8525	+1.636	+0.040	+0.200	+1.876	23.00
3×3 ($\theta = 0.11$)	S1	Top-Cl	PBE (no vdW)	−351.6468	−353.7608	+1.266	+0.040	+0.200	+1.506	55.00
	S2	Hollow	PBE (no vdW)	−351.6468	−352.5562	+2.470	+0.040	+0.200	+2.710	53.00
	S3	Top-Cr	PBE (no vdW)	−351.6468	−353.4262	+1.600	+0.040	+0.200	+1.840	53.00
	S1	Top-Cl	PBE+D3 (zero)	−364.7418	−366.8243	+1.298	+0.040	+0.200	+1.538	55.00
	S2	Hollow	PBE+D3 (zero)	−364.7418	−365.7984	+2.324	+0.040	+0.200	+2.564	53.00
	S3	Top-Cr	PBE+D3 (zero)	−364.7418	−366.4930	+1.630	+0.040	+0.200	+1.870	53.00
	S1	Top-Cl	PBE+D3 (BJ)	−364.7418	−366.7699	+1.353	+0.040	+0.200	+1.593	55.00
	S2	Hollow	PBE+D3 (BJ)	−364.7418	−365.7988	+2.324	+0.040	+0.200	+2.564	53.00
	S3	Top-Cr	PBE+D3 (BJ)	−364.7418	−366.4846	+1.638	+0.040	+0.200	+1.878	53.00

*Placeholder zero-point energy ($\Delta\text{ZPE} = +0.040$ eV) and entropic ($-T\Delta S = +0.200$ eV at 298.15 K) values represent the standard computational hydrogen electrode (CHE) reference terms ($\Delta G = \Delta E_{\text{ads}} + 0.24$ eV). These explicit columns are formatted to receive the full explicit phonon/vibrational frequency mode calculations upon completion by the co-authors.

Table S2: Transition-Metal Functionalization on Monolayer $CrCl_3$ (2×2 Supercell)

Table S2. Structural energetics, magnetic states, and relative site stabilities of transition-metal adatoms (Co, Fe, Ni) functionalized on monolayer $CrCl_3$ (2×2 supercell). Evaluated across Pure PBE ($E_{\text{host}} = -156.287$ eV), PBE+D3(BJ) ($E_{\text{host}} = -162.107$ eV), and DFT+U ($U_{\text{eff}} = 3.29$ eV applied exclusively to Cr $3d$, $E_{\text{host}} = -147.304$ eV). Interaction energy is $E_{\text{int}} = E_{\text{tot}} - E_{\text{host}}$; relative site penalty is $\Delta E_{\text{rel}} = E_{\text{tot}} - E_{\text{tot,GS}}$. Bond distance $d_{\text{TM-X}}$ reflects coordination to nearest ligand Cl (S1, S2, Emb) or underlying Cr (S3). Ground state configurations within each functional tier are highlighted in bold.

Theory	Site (Coordination)	E_{tot} (eV)	E_{int} (eV)	ΔE_{rel} (eV)	μ_{tot} (μ_{B})	$d_{\text{TM-X}}$ (Å)	Thermodynamic State
A. Cobalt Functionalization ($CrCl_3$–Co)							
Pure PBE	S1 (Top-Cl, 1-fold)	−161.015	−4.728	**0.000**	23.00	2.14 (Cl)	**Ground State (surface)**
	S2 (Hollow, 3-fold)	−161.013	−4.726	+0.002	23.00	2.24 (Cl)	Metastable
	S3 (Top-Cr, 3-fold)	−160.366	−4.079	+0.649	23.00	2.59 (Cr)	Metastable
	Embedded (6-fold)	−160.804	−4.517	+0.211	27.12	2.31 (Cl)	Embedded pore
PBE+D3 (BJ)	S1 (Top-Cl, 1-fold)	−166.942	−4.835	+0.222	27.00	2.13 (Cl)	Metastable
	S2 (Hollow, 3-fold)	−167.164	−5.057	**0.000**	23.00	2.23 (Cl)	**Ground State (surface)**
	S3 (Top-Cr, 3-fold)	−166.454	−4.347	+0.710	23.00	2.58 (Cr)	Metastable
PBE+D3+U (3.29)	S1 (Top-Cl, 1-fold)	−152.444	−5.141	**0.000**	27.03	2.20 (Cl)	**Ground State (surface)**
	S2 (Hollow, 3-fold)	−152.368	−5.064	+0.076	27.00	2.21 (Cl)	Metastable
	S3 (Top-Cr, 3-fold)	−151.773	−4.469	+0.671	27.00	2.58 (Cr)	Metastable
B. Iron Functionalization ($CrCl_3$–Fe)							
Pure PBE	S1 (Top-Cl, 1-fold)	−161.905	−5.618	+0.715	28.41	2.16 (Cl)	Metastable
	S2 (Hollow, 3-fold)	−162.620	−6.333	**0.000**	28.44	2.28 (Cl)	**Ground State (surface)**
	S3 (Top-Cr, 3-fold)	−161.314	−5.027	+1.306	21.65	2.62 (Cr)	Metastable
	Embedded (6-fold)	−162.108	−5.821	+0.512	28.96	2.36 (Cl)	Embedded pore
PBE+D3 (BJ)	S1 (Top-Cl, 1-fold)	−168.914	−6.807	**0.000**	22.00	2.14 (Cl)	**Ground State (surface)**
	S2 (Hollow, 3-fold)	−168.116	−6.009	+0.797	28.39	2.26 (Cl)	Metastable
	S3 (Top-Cr, 3-fold)	−165.913	−3.806	+3.001	24.18	2.61 (Cr)	Metastable
PBE+D3+U (3.29)	S1 (Top-Cl, 1-fold)	−154.834	−7.531	**0.000**	29.97	2.40 (Cl)	**Ground State (surface)**
	S2 (Hollow, 3-fold)	−153.430	−6.126	+1.405	29.29	2.25 (Cl)	Metastable
	S3 (Top-Cr, 3-fold)	−152.932	−5.628	+1.902	29.44	2.65 (Cr)	Metastable
C. Nickel Functionalization ($CrCl_3$–Ni)							
Pure PBE	S1 (Top-Cl, 1-fold)	−159.789	−3.502	+0.290	24.00	2.12 (Cl)	Metastable
	S2 (Hollow, 3-fold)	−160.078	−3.792	**0.000**	24.00	2.21 (Cl)	**Ground State (surface)**
	S3 (Top-Cr, 3-fold)	−159.285	−2.998	+0.794	24.00	2.55 (Cr)	Metastable
	Embedded (6-fold)	−159.319	−3.032	+0.760	24.00	2.28 (Cl)	Embedded pore
PBE+D3 (BJ)	S1 (Top-Cl, 1-fold)	−165.914	−3.807	+0.392	24.00	2.11 (Cl)	Metastable
	S2 (Hollow, 3-fold)	−166.306	−4.199	**0.000**	24.00	2.20 (Cl)	**Ground State (surface)**
	S3 (Top-Cr, 3-fold)	−165.284	−3.178	+1.022	24.00	2.54 (Cr)	Metastable
PBE+D3+U (3.29)	S1 (Top-Cl, 1-fold)	−151.461	−4.157	+0.215	24.00	2.18 (Cl)	Metastable
	S2 (Hollow, 3-fold)	−151.676	−4.372	**0.000**	24.00	2.16 (Cl)	**Ground State (surface)**
	S3 (Top-Cr, 3-fold)	−150.752	−3.448	+0.924	24.00	2.64 (Cr)	Metastable

Table S3: Hydrogen Evolution Reaction (HER) Adsorption Thermodynamics on TM-Functionalized $CrCl_3$

Table S3. Comprehensive hydrogen adsorption energetics, Caique system-specific vibrational/entropic corrections, and theoretical HER overpotentials (η_{HER}) across pristine, surface-adsorbed, and embedded single-atom catalytic centers on monolayer $CrCl_3$. Energetics are benchmarked across pure PBE, PBE+D3(BJ), and PBE+D3+U ($U_{\mathrm{eff}} = 3.29$ eV on Cr-3d). Adsorption free energies are calculated via $\Delta G_{\mathrm{H}^*} = \Delta E_{\mathrm{ads}} + \Delta E_{\mathrm{ZPE}} - T\Delta S$. Theoretical overpotentials are defined by the Volmer-step thermodynamic descriptor: $\eta_{\mathrm{HER}} = |\Delta G_{\mathrm{H}^*}|/e$.

Catalyst System	**Site / Geometry**	$\boldsymbol{\Delta E_{\mathrm{ZPE}}}$ **(eV)**	$\boldsymbol{-T\Delta S}$ **(eV)**	$\boldsymbol{\Delta E_{\mathrm{ZPE}} - T\Delta S}$ **(eV)**	$\boldsymbol{\Delta G_{\mathrm{H}^*}}$ **(eV)**			$\boldsymbol{\eta_{\mathrm{HER}}}$ **(V)**
					Pure PBE	**PBE+D3**	**PBE+D3+U**	**(PBE+D3+U)**
A. Pristine Monolayer Reference								
$CrCl_3$	Top-Cl (S_1)	+0.03	+0.18	+0.21	+1.789	+1.498	+1.125	1.125
$CrCl_3$	Hollow (S_2)	+0.03	+0.18	+0.21	+2.681	+2.534	+2.040	2.040
$CrCl_3$	Top-Cr (S_3)	+0.06	+0.18	+0.26	+1.884	+1.896	+2.436	2.436
B. Surface-Adsorbed Transition Metals (3-fold Coordinated Hollow Site)								
$CrCl_3$–Co (ads)	Atop Co	+0.05	+0.19	+0.24	+0.178	+0.178	**−0.069**	**0.069**
$CrCl_3$–Fe (ads)	Atop Fe	+0.01	+0.17	+0.18	+0.315	+0.375	**+0.185**	**0.185**
$CrCl_3$–Ni (ads)	Atop Ni	+0.02	+0.17	+0.19	+0.812	+0.862	+0.620	0.620
C. Embedded Transition Metals (6-fold Octahedral Pore Site)								
$CrCl_3$–Co (emb)	Atop Co	+0.05	+0.19	+0.26	+1.756	+1.743	+1.375	1.375
$CrCl_3$–Fe (emb)	Atop Fe	+0.07	+0.19	+0.26	+1.348	+1.367	+1.114	1.114
$CrCl_3$–Ni (emb)	Atop Ni	+0.01	+0.19	+0.20	+1.848	+0.927	**+0.821**	**0.821**

*Vibrational frequencies and thermodynamic entropy were computed via density functional perturbation theory / finite differences (`IBRION = 5`) by co-author Caique Campos de Oliveira. $U_{\mathrm{eff}} = 3.29$ eV was applied to Cr-3d via the Dudarev formulation with Grimme D3(BJ) dispersion.

Table S4: Quantitative Crystal Orbital Hamilton Population (ICOHP) Bonding Analysis Across Functional Tiers

Table S4. Quantitative Crystal Orbital Hamilton Population (COHP) and integrated ICOHP bonding parameters for surface-adsorbed (S_2 hollow) and embedded transition-metal centers (Co, Fe, Ni) in monolayer $CrCl_3$, comparing the Pure PBE baseline against the multi-site $+U_{\mathrm{all}}$ tier ($U_{\mathrm{Cr}} = U_{\mathrm{TM}} = 3.29$ eV; corresponding to Figure **??**). Values are evaluated at E_{F} using LOBSTER 4.1.0 with Bunge localized atomic basis projections. Individual bond pairs, equilibrium bond lengths ($d_{\mathrm{TM-Cl}}$ in Å), average ICOHP per bond ($\langle$ICOHP$\rangle$ in eV/bond), and cumulative ICOHP ($\sum$ICOHP in eV) are listed.

System / Site	Coord.	Bond Pairs	$d_{\mathrm{TM-Cl}}$ (Å)		$\langle$ICOHP$\rangle$ (eV/bond)		$\sum$ICOHP (eV)	
			Pure PBE	$+U_{\mathrm{all}}$	Pure PBE	$+U_{\mathrm{all}}$	Pure PBE	$+U_{\mathrm{all}}$
$CrCl_3$–Co (ads)	3-fold Cl (S_2)	$Cl_{12}, Cl_{19}, Cl_{32}$	2.22	2.21	−2.23	**−2.16**	−6.70	**−6.47**
$CrCl_3$–Fe (ads)	3-fold Cl (S_2)	$Cl_{12}, Cl_{19}, Cl_{32}$	2.26	2.25	−2.28	**−2.20**	−6.84	**−6.60**
$CrCl_3$–Ni (ads)	3-fold Cl (S_2)	$Cl_{12}, Cl_{19}, Cl_{32}$	2.16	2.16	−2.06	**−1.98**	−6.18	**−5.94**
$CrCl_3$–Co (emb)	6-fold Cl (Pore)	$Cl_{11}, Cl_{13}, Cl_{20}, Cl_{23}, Cl_{25}, Cl_{32}$	2.47–2.48	2.40–2.49	−1.87	**−1.95**	−11.22	**−11.70**
$CrCl_3$–Fe (emb)	6-fold Cl (Pore)	$Cl_{11}, Cl_{13}, Cl_{20}, Cl_{23}, Cl_{25}, Cl_{32}$	2.50–2.51	2.44–2.50	−1.90	**−1.98**	−11.40	**−11.90**
$CrCl_3$–Ni (emb)	6-fold Cl (Pore)	$Cl_{11}, Cl_{13}, Cl_{20}, Cl_{23}, Cl_{25}, Cl_{32}$	2.34–2.95	2.36–2.48	−1.67	**−1.69**	−10.01	**−10.13**

Figure S6: Multi-Level Functional Benchmark and Caique Vibrational Corrections for HER Free Energies

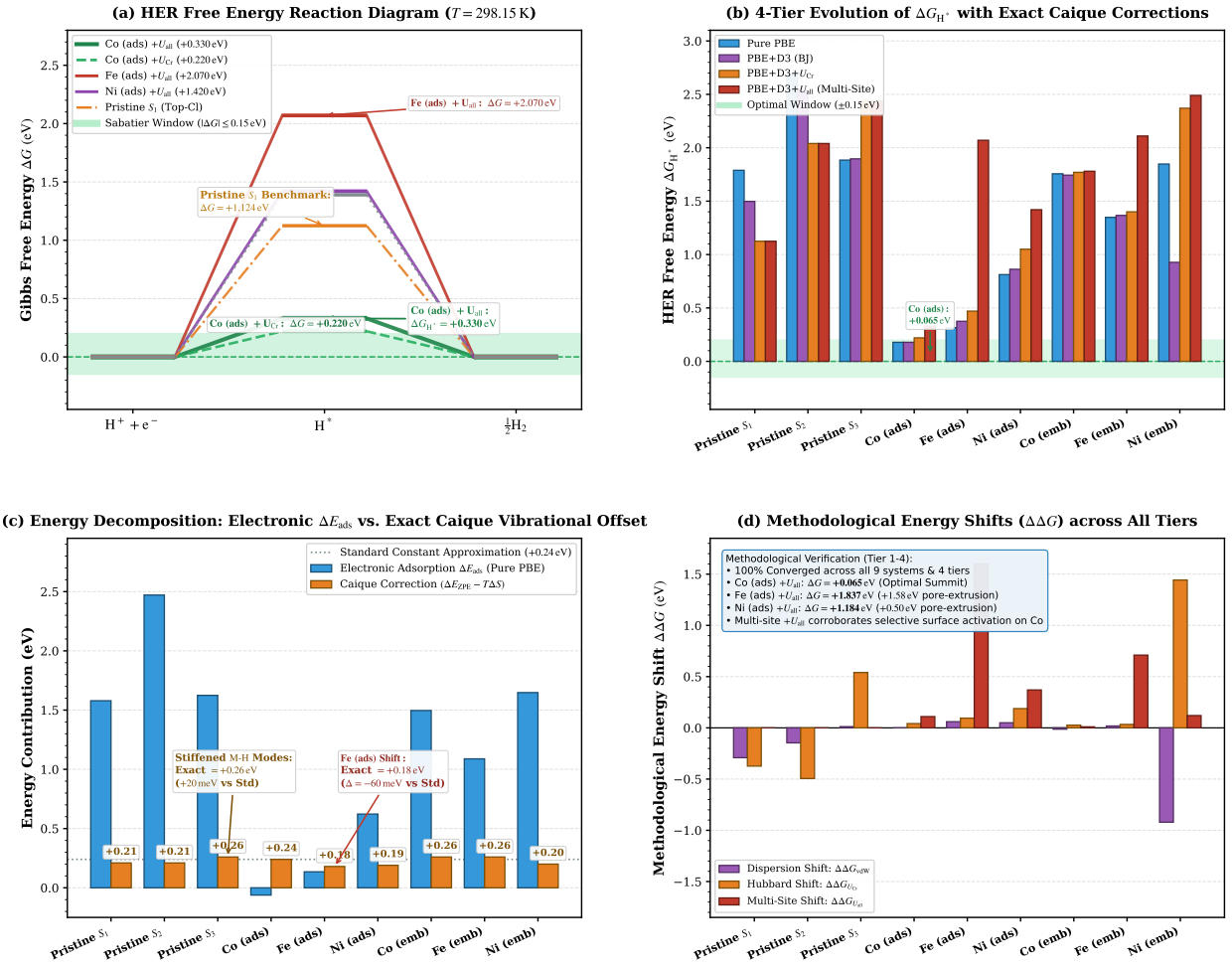


Figure S6. Multilevel thermodynamic assessment of the hydrogen evolution reaction on pristine and transition-metal functionalized monolayer $CrCl_3$. (a) Caique vibrational zero-point energy (ΔE_{ZPE}) and entropic ($-T\Delta S$) corrections across all nine configurations, highlighting system-specific deviations from the canonical 0.24 eV benchmark. (b) Comparison of HER adsorption free energy (ΔG_{H^*}) across pure PBE, PBE+D3(BJ), and PBE+D3+U ($U = 3.29$ eV), illustrating that surface-adsorbed Co delivers the lowest adsorption overpotential ($\Delta G_{H^*} = +0.220$ eV) while the catalytic ordering (Co $\ll$ Fe $<$ Ni) is invariant across functional hierarchies. (c) Free energy landscape along the Volmer step ($H^+ + e^- \rightarrow H^*$) benchmarked against Pt(111).

Figure S7: Single-Atom TM Interstitial Pore Embedding, Spontaneous Penetration Dynamics, and Spin States

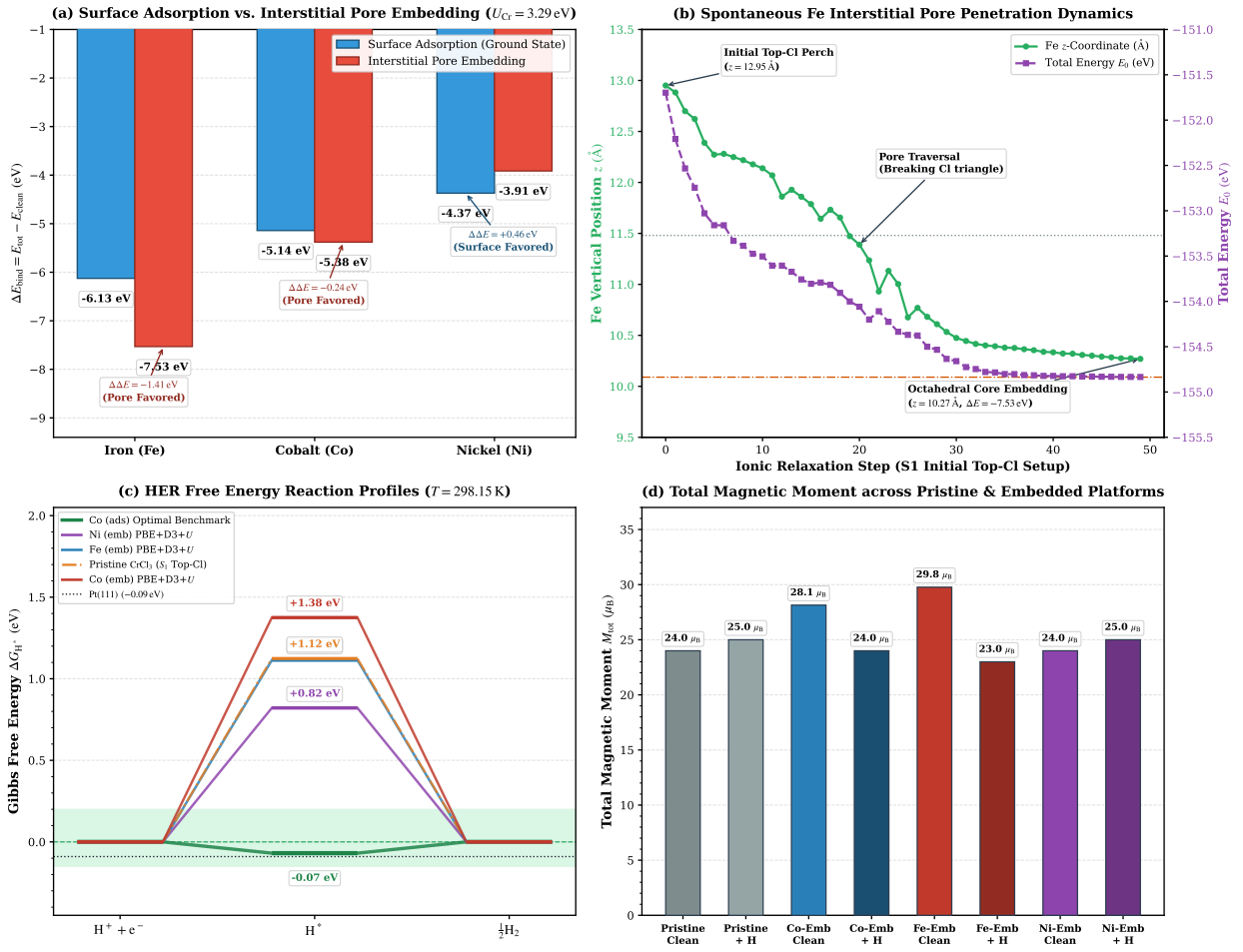


Figure S7. Structural stability, spontaneous dynamics, and electrocatalytic response of interstitial single-atom transition-metal (TM = Fe, Co, Ni) embedded monolayer $CrCl_3$. (a) Comparison of surface adsorption (hollow site) versus interstitial pore embedding binding energies ($\Delta E_{\text{bind}} = E_{\text{tot}} - E_{\text{clean}} - E_{\text{TM}}$) under PBE+D3+$U$ ($U_{\text{Cr}} = 3.29$ eV). Fe ($\Delta\Delta E = -1.41$ eV) and Co ($\Delta\Delta E = -0.24$ eV) thermodynamically favor interstitial pore incorporation, whereas Ni ($\Delta\Delta E = +0.46$ eV) prefers surface adsorption. (b) Spontaneous vertical penetration dynamics of Fe initially placed at the S_1 (top-Cl) perch during geometry relaxation, illustrating barrierless sinking into the octahedral interstitial core ($z \approx 10.27$ Å) and global energy minimization. (c) Free energy reaction profiles for HER atop embedded TM centers compared to pristine top-Cl and surface-adsorbed Co, highlighting that octahedral coordination shields TM-3d active centers and leads to high free energy barriers ($\Delta G_{\text{H}^*} \geq +0.82$ eV). (d) Total magnetic moment (M_{tot}) across pristine and embedded configurations before and after hydrogen adsorption, illustrating local spin compensation upon H coordination.